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Research and Applications

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Discover important donor-recipient risk factors and interactions in heart transplant primary graft dysfunction with machine learning

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Abstract

Objectives: Primary graft dysfunction (PGD) is an essential outcome after the heart transplant, which causes severe complications and symptoms for recipients. The in advance prediction of PGD can help the transplant physician better manage the risks of PGD occurrence for patients. Domain experts have identified some important risk factors leading to PGD. However, a widely accepted PGD prediction method is lacking from a computational perspective. In this work, we focus on the prediction of PGD after heart transplant with machine learning (ML).

Materials and Methods: With the strong power of artificial intelligence, we propose to design a ML algorithm to precisely predict the PGD with the donor and recipient features. Moreover, we apply the computational method to automatically identify important features and interactions between them.

Results: To evaluate the effectiveness of the ML algorithm in PGD prediction, we curated a PGD patients' cohort from the United Network for Organ Sharing database, which contains 8008 recipients. 5 commonly used ML models are used for performance comparison. The multi-layer perceptron model achieves superior performance, as measured by area under the receiver operating characteristic curve (AUROC), at 0.868. We identify the top 20 important features and interactions between donors and recipients. Clinical analyses are conducted on the identified features and interactions.

Discussion: We summarize the contributions of this work from three aspects including methodology, clinical analysis, and insights. We discuss the limitations of this work on data, model, and real-world implementation perspectives. Additionally, we further discuss the future directions to extend this work to more organ types and diseases.

Conclusion: In summary, ML has promising applications in PGD prediction as a computational tool for clinical study. We can also use the ML model to help us identify and discover new risk factors and interactions between donor and recipient.

Key words: heart transplant; primary graft dysfunction; machine learning; explainable artificial intelligence.

Introduction

In 2014, the International Society of Heart and Lung Transplantation (ISHLT) established standardized guidelines for diagnosing, managing, and treating Primary Graft Dysfunction (PGD) during its annual meeting.[1] These guidelines outline criteria and risk factors for identifying PGD post-heart transplant, encompassing factors related to the donor, recipient, and surgical procedures. PGD is defined by abnormal cardiac performance without other causative factors. While ISHLT's standards serve as a foundational framework, research from various regions and institutions has further evaluated and refined these criteria. For instance, Singh et al. assessed PGD incidence in the UK using ISHLT criteria.[2] At the same time, Ahsan et al. introduced a risk-scoring method for predicting PGD in orthotopic heart transplants, leveraging individual predictors such as milrinone use and donor pH.[3]

While many existing approaches to predicting PGD rely on heuristic methods based on expert-defined rules,[4] these approaches often face challenges when applied to diverse patient populations across different regions and timeframes.[5] Furthermore, the accuracy of rule-based methods may not reach optimal levels. With the emergence of artificial intelligence (AI), there's a promising avenue for more precise PGD prediction. Yet, there's a noticeable gap in the development of computational methods leveraging AI and machine learning (ML) for PGD prediction. Notable efforts include Michelson et al.'s ML algorithm for predicting PGD in lung transplants[6] and a ML-based approach for heart transplant PGD

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prediction.[7] Nevertheless, a comprehensive comparison between ML and traditional heuristic methods, along with the novel insights provided by ML, remains largely unexplored.

Moreover, understanding feature attribution[8] and interactions[9] between donors and recipients is crucial for deciphering the mechanisms behind PGD in heart transplants and pinpointing its risk factors. While most studies rely on domain expertise to analyze PGD risk factors,[4,5] there's a notable lack of research leveraging computational tools to systematically and exhaustively uncover new risk factors and their interplay. For instance, manually enumerating feature combinations becomes impractical, with 1770 potential interactions among just 60 features. The intricacy of these interactions presents a significant challenge, with only limited efforts currently aiming to unearth these valuable feature interactions. Future research leveraging advanced computational methods could provide deeper insights into the complex interactions influencing PGD.

Therefore, we designed this study to develop a computational approach for PGD prediction, leveraging the advancements in ML and focusing on discerning feature attributions and interactions within each sample. We hope the results from this study will aid clinicians in making informed decisions regarding heart organ allocation.

Background and significance

Primary graft dysfunction poses a significant challenge in heart transplant outcomes.[10] This condition presents as a severe and acute dysfunction of the transplanted heart within the first 24 hours post-transplant, often leading to significant morbidity and mortality and becoming a primary contributor to graft failure.[11] Early and accurate diagnosis of PGD is crucial for optimizing heart transplant outcomes. Symptoms can range from mild cardiac dysfunction to severe cardiogenic shock.[12] Furthermore, pre-transplant diagnosis of PGD offers detailed insights into graft viability, aiding clinicians in making informed decisions regarding heart organ allocation.

Materials and methods

Data preparation

Data source

Data were obtained from the United Network for Organ Sharing (UNOS) database, which collects all data related to waiting list registrations and transplantation that have been listed or performed in the United States and reported to the Organ Procurement and Transplantation Network (OPTN) since October 1, 1987. Extensive baseline data were collected for each patient, including demographics, clinical and laboratory details at the time of listing and transplantation, and follow-up survival data.

Cohort and feature selection

First, data were extracted from the UNOS database and preprocessed to obtain the study dataset using the higher-level programming language Python. We excluded patients with multiple transplants during the index hospitalization. We filtered out features with the expert's domain knowledge and features with a higher than 40% missing rate. We also generate several pairs of features of blood type, age, gender, and race between donor and recipient. There are 66 features selected and created for the heart transplant PGD prediction. Details of features are provided in Tables S1 and S2.

Data preprocessing

There are categorical and numerical features in the input data. We encode the categorical features into numerical ones for homogeneity. For the missing data, we apply the kNN imputer, which calculates the similarity between patients, and use the value from the nearest patient as imputation. We chose kNN for missing data imputation due to its simplicity, effectiveness, and suitability for our dataset's structure. kNN provided an efficient solution.

Problem formulation

The heart transplant PGD cohort can be represented as $= \{(r_i,\ o_i,\ y_i)\}_{i=1}^{N}$, where $r_i \in R^{M_r}$ denotes the recipient's features (eg, recipient height, weight, age, etc), $o_i \in R^{M_o}$ denotes the organ's features (eg, donor blood type, cause of death, etc). The $y \in \{0,\ 1\}$ indicates the diagnosis of PGD after a heart transplant. M_r and M_o are the feature dimensions, respectively, and N is the number of patients in the cohort. The goal is to train a ML model f to precisely predict the y_i with the $\mathbf{r}_i$ and $\mathbf{o}_i$ as input. Furthermore, we aim to calculate the attributions $\mathbf{a}$ and interactions $\mathbf{b}$ of each feature with the well-learned f so that we can interpret the predictions of the PGD.

PGD prediction model

To precisely predict the PGD, we propose a ML algorithm, as presented in Figure 1, to take advantage of the donor's and patient's data.

The algorithm consists of several steps, including patient cohort selection, feature selection, train-test data split, model training, and prediction explanation. In this algorithm, we implement and apply a multi-layer perceptron (MLP) for the

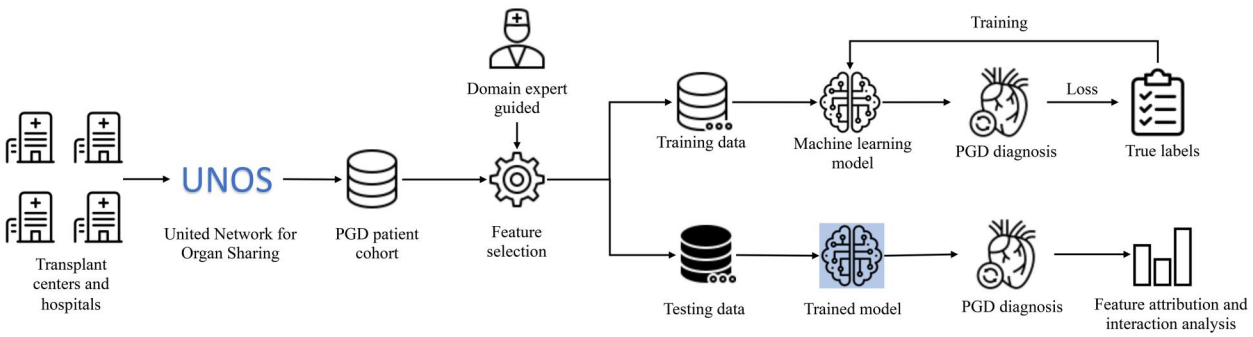

Figure 1. The ML algorithm to predict the PGD in heart transplant.

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model prediction. To evaluate the performance of different ML models, we compare with a linear model (logistic regression), a tree model (GBDT, Decision tree, LGBM), and others, for example, support vector machine (SVM). The predictive accuracy of the ML models was evaluated by measuring the total area under the curve (AUC) of the receiver-operator curve (ROC) and the area under the precision-recall curve (AUPRC), which are commonly used metrics in the classification task, especially when the size difference between PGD patients and non-PGD patients are significant.

Salient and interactive feature discovery

Single feature importance

We adopt the method from[13] to calculate the attribution of each feature to the predictions. The Integrated Gradient CITE method is applied to estimate the Aumann-Shapley value. The attribution for each feature can be calculated as follows:

$$a_p(x) = \left(x_p - x_p^{'}\right) \times \int_{\alpha=0}^{1} \frac{\delta f(x^{'} + \alpha(x - x^{'}))}{\delta x_p} d\alpha \qquad (1)$$

where x is the data sample and p is the pth feature of x to be explained, x' is the baseline value. $p(x)$ is the attribution of the pth feature in x.

Feature interaction identification

Based on the integrated gradients, we can calculate the influence of feature q on feature p as follows:

$$\beta_{p,q}(x) = \alpha_q\,(\alpha_p(x)), \qquad (2)$$

where $\beta_{p,\,q}(x)$ is the interaction of the qth feature on pth feature in x. This value is symmetrical indicating the interaction of feature q with p is identical to feature p with q as follows.

$$\beta_{p,\,q}(x) = \beta_{q,\,p}(x) \qquad (3)$$

Results

Patient cohort and incidence of PGD

We selected 8008 patients who received heart transplants registered in the UNOS database. 2146 patients (26.8%) are diagnosed with PGD, and 5862 (73.2%) patients have successful grafts. The patients were selected in the period of 1987-2018. Among the patients, 5671 are male and 2337 are female. There are 5 categories of ethnicity in the patients; 5304 are white, 1613 are African-American, 242 are Asian, and 120 are other ethnicities. The detailed information of the patients is presented in Table 1. We categorized the selected features for donors and patients into 4 kinds, including demographic, medical history, lab tests, and others; the detailed features information is presented in Table 2.

PGD prediction performance

The prediction results are presented in Table 3. Additionally, the ROC curve and PRC curve are presented in Figure 2. We analyze the results and draw insights as follows.

MLP achieves significantly better performance than other baselines. From Table 3, we can see that the MLP has superior prediction accuracy compared to other baselines. For example, the MLP surpasses the LightGBM by 7.6% on AUROC and 16.3% on AUPRC. This may be due to MLP's ability to directly learn complex, non-linear patterns from the data without extensive feature

Table 1. Demographic statistics of patient cohort.

	Total (8008)	PGD patients (2146, %)	Non-PGD patients (5862, %)
Recipient characteristics			
Age, median (IQR)	52 (33-61)	51 (32-59)	53 (34-62)
Gender: Male, %	5671 (70.8)	1532 (71.4)	4139 (70.6)
Gender: Female, %	2337 (29.2)	614 (28.6)	1723 (29.4)
Race: White	5304 (66.2)	1534 (71.5)	3770 (64.3)
Race: Black	1613 (20.1)	413 (19.2)	1200 (20.5)
Race: Hispanic	729 (9.1)	146 (6.8)	583 (9.9)
Race: Asian	242 (3.0)	33 (1.5)	209 (3.6)
Race: Others	120 (1.5)	20 (0.9)	100 (1.7)
BMI:	25.6 (21.6-29.5)	25.1 (21.4-28.8)	25.8 (21.8-29.8)
Blood type: A	3209 (40.1)	912 (42.5)	2297 (39.2)
Blood type: B	1125 (14.0)	279 (13.0)	846 (14.4)
Blood type: O	3226 (40.3)	847 (39.5)	2379 (40.6)
Blood type: AB	448 (5.6)	108 (5.0)	340 (5.8)
Donor characteristics			
Age, median (IQR)	27 (19-39)	27 (18-41)	27 (19-38)
Gender: Male	5411 (67.6)	1405 (65.5)	4006 (68.3)
Gender: Female	2597 (32.4)	741 (34.5)	1856 (31.7)
Race: White	5162 (64.5)	1530 (71.3)	3632 (62.0)
Race: Black	1289 (16.1)	300 (14.0)	1028 (17.5)
Race: Hispanic	1282 (16.0)	254 (11.8)	989 (16.9)
Race: Asian	143 (1.8)	29 (1.4)	114 (1.9)
Race: Others	132 (1.6)	33 (1.5)	99 (1.7)
BMI:	24.9 (21.3-28.8)	23.7 (19.8-27.2)	25.4 (21.8-29.4)
Blood type: A	2819 (35.2)	773 (36.0)	2046 (34.9)
Blood type: B	832 (10.4)	211 (9.8)	621 (10.6)
Blood type: O	4181 (52.2)	1127 (52.5)	3054 (52.1)
Blood type: AB	174 (2.2)	33 (1.5)	141 (2.4)

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Table 2. Characteristics of heart transplant (the details for each feature are presented in Table S2).

	Demographic	Medical history	Lab tests
Recipient	GENDER, ABO, CITIZENSHIP, ETHCAT, INIT_HGT_CM_CALC, INIT_WGT_KG_CALC, AGE, BMI_CALC, BSA	NUM_PREV_TX, DIAB, CEREB_VASC, INOTROP_VASO_SYS_TCR, INOTROP_VASO_MN_TCR, PRIOR_CARD_TCR, INFECT_IV_DRUG_TRR, INOTROP_VASO_CO_TRR, INOTROP_VASO_MN_TRR, INOTROP_VASO_SYS_TRR, PRIOR_LUNG_SURG_TRR, PREV_TX, PREV_TX_ANY, DIABETES_DON, PRIOR_CARD_SURG_TRR	HEMO_SYS_TCR, HEMO_PA_DIA_TCR, HEMO_PA_MN_TCR, HEMO_PCW_TCR, HEMO_CO_TCR, CREAT_TRR,
Donor	AGE_DON, ETHCAT_DON, ABO_DON, GENDER_DON, HGT_CM_DON_CALC, WGT_KG_DON_CALC, BMI_DON_CALC, BSA_DON	HIST_COCAINE_DON, HIST_DIABETES_DON, HIST_OTH_DRUG_DON, COD_CAD_DON, HIST_MI	PT_T4_DON, PULM_INF_DON, DDAVP_DON, LV_EJECT
Others			
Operations	STEROID, ISCHTIME, DISTANCE, PROC_TY_HR, LV_EJECT_METH		
Pair features	CRSMATCH_DONE, PAIR_GENDER, PAIR_RACE, PAIR_AGE, PAIR_BMI, PAIR_ABO, PAIR_BSA		
Life support	ECMO_TCR, OTH_LIFE_SUP_TCR, LIFE_SUP_TCR, ECMO_TRR, LABP_TRR, VENT_SUPPORT_TRR, VENT_SUPPORT_AFTER_LIST		

Table 3. Prediction performance compared to other baseline ML models. The bold values indicate the best performance of ML models.

Model	Model category	AUROC	AUPRC
LR	Linear model	$0.731 + 0.004$	$0.545 + 0.010$
SVM	Others	$0.525 + 0.004$	$0.291 + 0.008$
GBDT	Tree model	$0.801 + 0.006$	$0.682 + 0.013$
Decision Tree	Tree model	$0.756 + 0.009$	$0.507 + 0.020$
LightGBM	Tree model	$0.807 + 0.006$	$0.683 + 0.016$
MLP	Neural Network	$\mathbf{0.868 + 0.006}$	$\mathbf{0.794 + 0.015}$

engineering. They handle data variability well and benefit from advancements in deep learning, making them more effective in the PGD predictive task.

SVM cannot precisely predict the PGD compared to other categories models. SVM shows the unsatisfying performance in Table 3, which has much lower AUROC and AUPRC than the other baselines. This is possibly caused by the SVM's reliance on linear separability assumptions, which are thus limited to the complex, non-linear data relationships typical in heart transplant patients' data. Additionally, the SVM may be sensitive to feature engineering and needs a specially designed feature engineering procedure.

The performance of ML-predicted PGD is robust with different patient distributions. Based on the 5-fold cross-validation, all the ML models are trained and tested on 5 different training and testing distributions. As shown in Table 3, the standard deviation among the 5-fold test is relatively small. Moreover, the curves of MLP on 5 folds are plotted in Figure 2, which shows the stability of the MLP's performance on different train-test distributions.

Feature importance ranking

The importance of each feature indicates the attribution of this feature to the prediction outcomes. We ranked the top 20 feature attributions in Figure 3 and listed the top 5 feature details in Table 4. Most of the top 20 features are important risk factors the physician will consider in their clinical practice. Meanwhile, some new features are discovered by the model and identified by the physicians as potential and new risk factors. Meanwhile, we have several findings through the scatter plot of the salient features in Figure 4. We summarize several findings as follows.

New risk factors that may affect the chance of PGD can be identified by feature importance ranking. For example, hemodynamics plays an essential role in the cause of PGD after a heart transplant (shown in Figure 3 as HEMO PA MN TCR, HEMO SYS TCR, etc). From Figure 4, a higher HEMO_PA_MN_TCR value can decrease the probability of PGD.[5] This may be because higher PA pressures can reflect enhanced organ perfusion and effective pre-transplant management, contributing to minimizing ischemia-reperfusion injury and improving post-transplant outcomes.

Larger age of the recipient or donor will both increase the risk of PGD after transplant. As shown in Figure 4, the higher the recipient's age is, the higher the chance of PGD contributed by the age. This finding also applied to the donor's age on the right side of the subfigure. The larger age of the recipient or donor increases the risk of PGD after heart transplantation due to age-related declines in cardiovascular function and resilience.[5] Older donors' hearts may be more prone to ischemia-reperfusion injury, while older recipients often have less robust cardiovascular systems and more comorbidities, complicating recovery. Consequently, the combined impact of aging on both donor heart quality and recipient recovery capacity heightens the risk of PGD.

Larger BMI of either recipient or donor puts the recipients at higher risk of PGD. As presented in the second row

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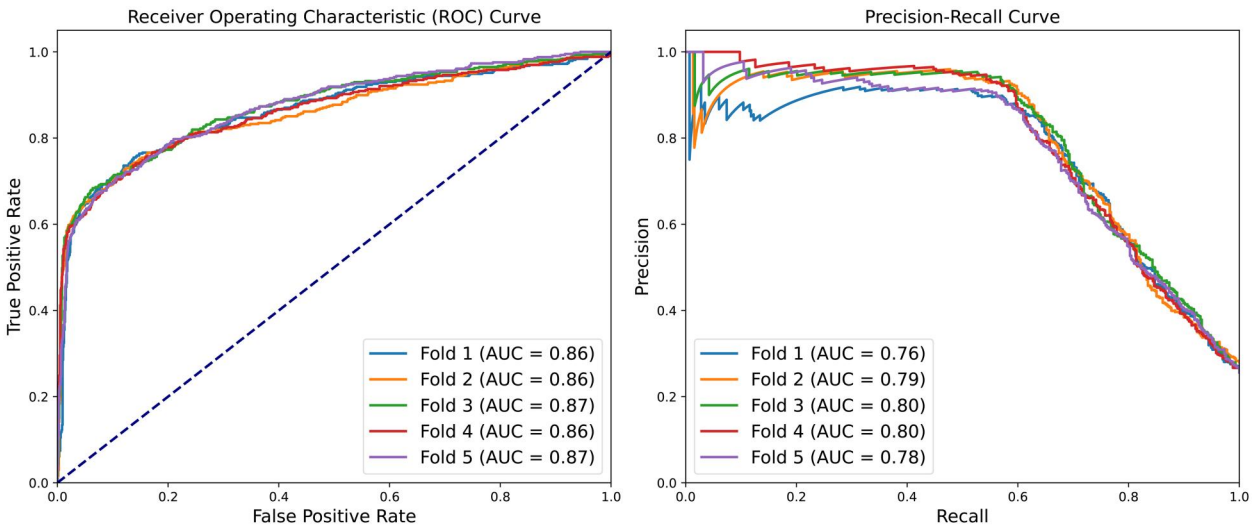

Figure 2. ROC and PRC curves for PGD prediction on 5-fold validation.

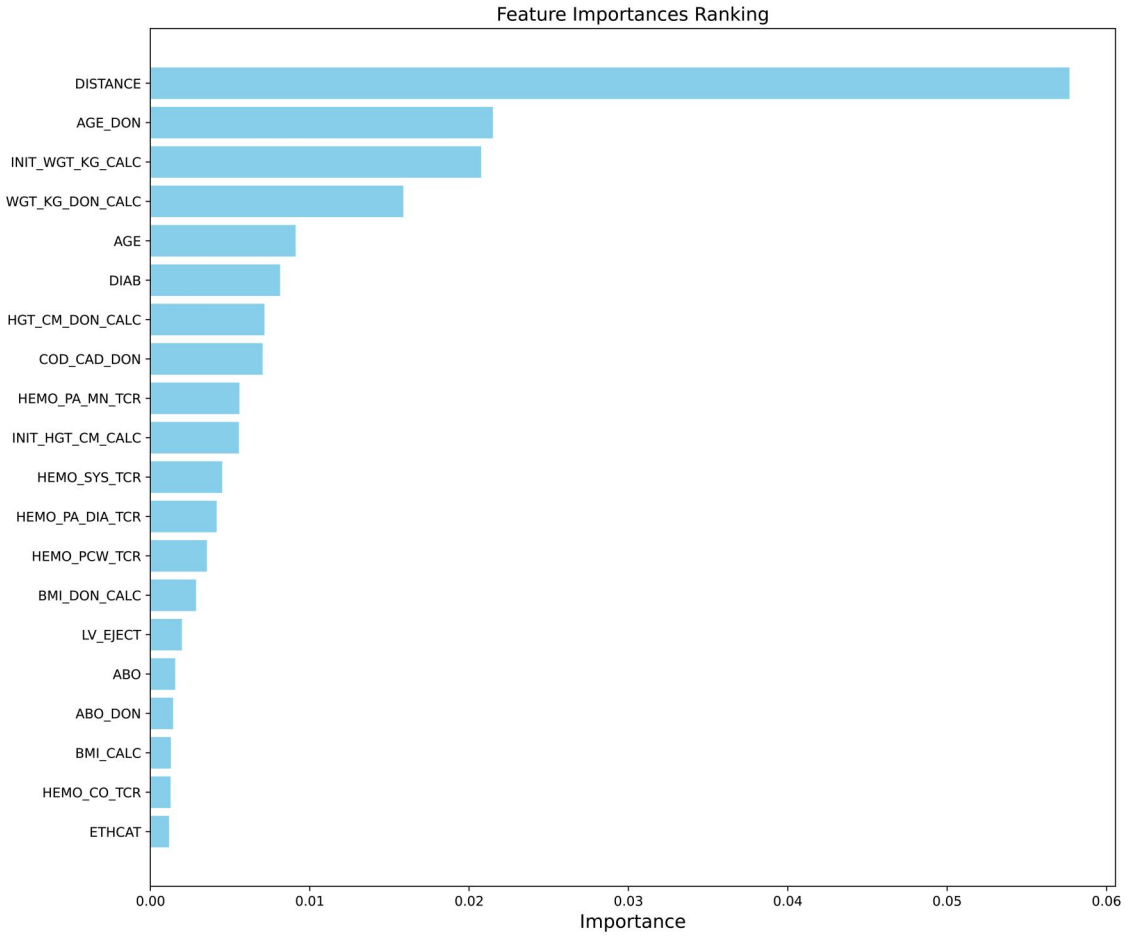

Figure 3. Single feature importance ranking.

of Figure 4, the higher BMI among recipients and donors will increase the PGD risk. A larger BMI in either the recipient or donor increases the risk of PGD because it is associated with comorbidities and impaired cardiac function.[14,15] For recipients, high BMI complicates recovery, while for donors, it can indicate poorer heart quality. Thus, elevated BMI in both parties heightens the risk of PGD.

Diabetes history of the recipient will increase the chance of PGD. We can also observe that diabetes among recipients will significantly increase the PGD after a heart transplant in Figure 4. A history of diabetes in the recipient

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Table 4. Summarization of PGD predictive single and combined features.

Importance ranking	Single features	Feature combinations
1	Distance between recipient and donor	Donor height and donor age
2	Donor age	Recipient's previous cardiovascular surgery history and previous lung surgery history
3	Recipient weight	Donor weight and recipient age
4	Donor weight	Donor age and recipient's previous lung surgery history
5	Recipient age	Recipient height and donor weight

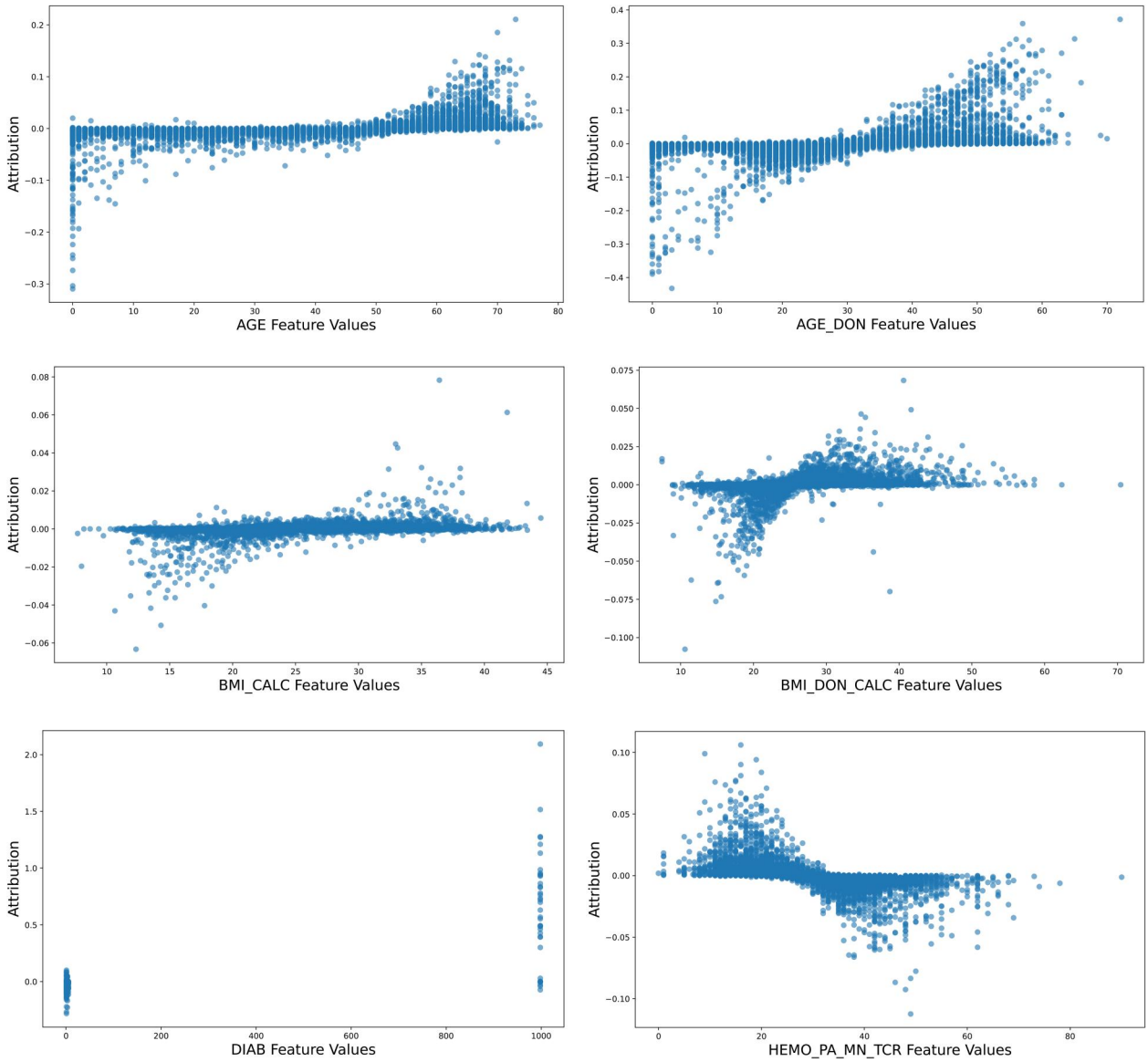

Figure 4. Scatter plot for selected salient features analysis.

increases the risk of PGD after heart transplantation due to several factors. Diabetes is associated with vascular complications, impaired wound healing, and increased susceptibility to infections, all of which can compromise the transplant recovery process. Additionally, diabetes can lead to poor glycemic control, further stressing the cardiovascular system and the newly transplanted heart.[16,17] These complications collectively elevate the risk of PGD in diabetic recipients.

Feature interactions discovery

We get the feature interactions from donors and patients. The global average interactions are calculated across all the patients. The top 20 ranking of feature interactions are shown in Figure 5. We can observe some feature interactions discovered by ML that correspond to human expertise. Moreover, we have found some new and insightful interactions. We will demonstrate them as follows with more details.

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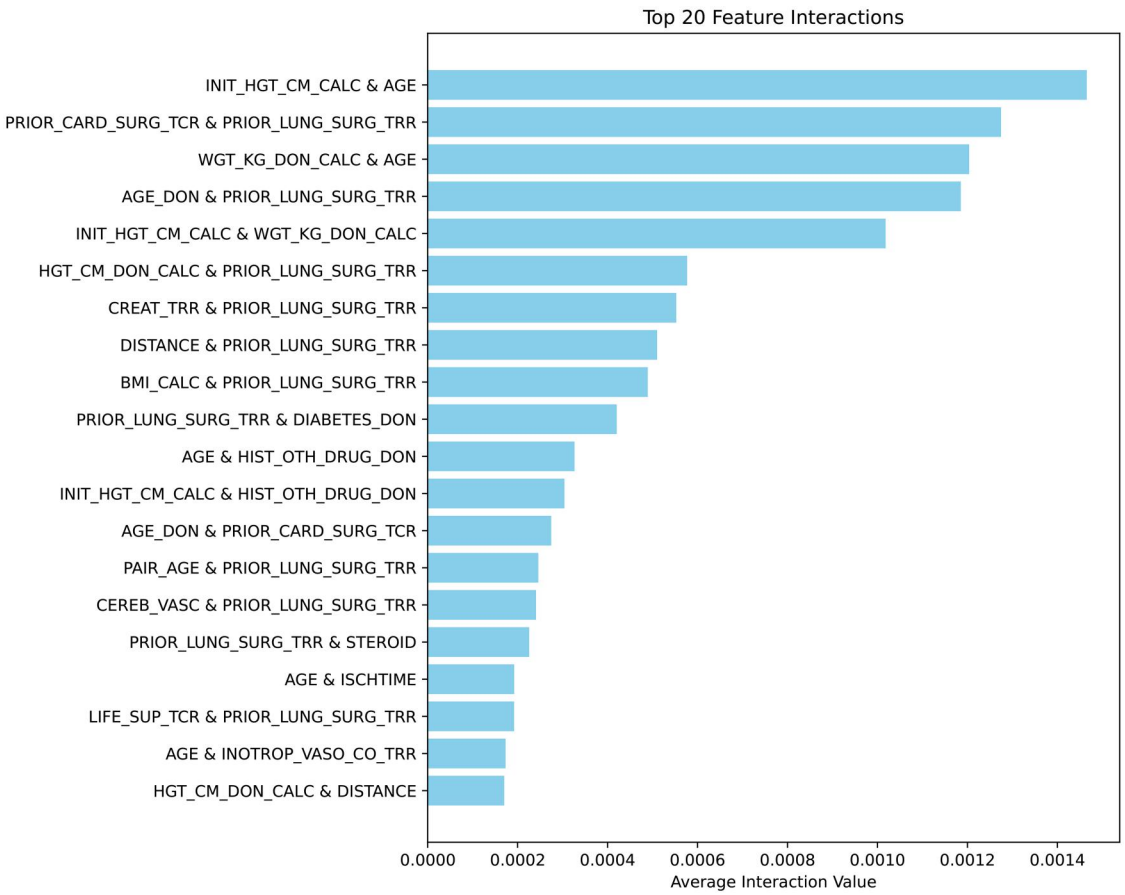

Figure 5. Feature interaction importance ranking.

The impact of feature interaction on the PGD is significantly smaller than single features. Compared to the results in Figures 3 and 5, the importance of single features is much more significant than the importance of feature interactions. For example, the importance of the single feature "DISTANCE" is nearly 0.06, much more significant than the importance of the "INIT_HGT_CM_CALC & AGE" interaction. This may be because individual features like recipient age, donor heart quality, BMI, and diabetes history directly and substantially affect the risk of PGD. These single features contribute to the physiological and medical conditions affecting transplant outcomes. In contrast, interactions between features tend to have a more complex and nuanced influence, which may not be as pronounced as the single features' direct effects. Therefore, while feature interactions can still provide valuable insights, their overall impact on PGD risk is generally less significant than individual features' primary effects.

While our analysis highlights the importance of individual features, the superior performance of deep learning models like MLPs stems from their ability to capture complex feature interactions. These interactions, which are often overlooked in traditional methods, play a critical role in improving predictive accuracy. Future research should focus on explicitly analyzing and interpreting these interactions to further enhance model performance and clinical relevance.

The age of the donor and recipient have important interactions with many other features. From Figure 5, the age of the recipient and donor interacts with 7 different features spanning from the height, weight of the recipient and donor, drug history of the recipient, surgery history, and other factors like ischemic time. These interactions arise because older recipients and donors often have more comorbid conditions, such as hypertension, diabetes, and atherosclerosis, which complicate transplantation and recovery. Older recipients may have a longer drug history affecting immune response,[18] and their surgery history can indicate previous procedures that complicate current outcomes. Additionally, older donor hearts are more susceptible to ischemia-reperfusion injury due to decreased regenerative capacity, making ischemic time a critical factor.[19,20] Understanding these complex interactions is essential for optimizing donor-recipient matching and improving post-transplant outcomes.

The previous surgery on lung or cardiovascular significantly interacts with other features. Figure 5 shows that the previous lung or cardiovascular surgery interacts with many other features, for example, age, height, creatine, etc The previous lung transplant has the effects of postoperative adhesions, reduced chest wall compliance, impaired respiratory function, and fibrotic changes, which can increase the risk of PGD.[21] The previous cardiology transplant had the effects of scar tissue formation, altered

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cardiac anatomy, impaired cardiac function, increased sensitivity to ischemia, and potential for increased inflammatory response.[21] These adverse effects have led to a higher risk of PGD.

Discussion

PGD is a significant contributor to perioperative morbidity and mortality and increases hospital length of stay and health care costs. A highly accurate risk prediction model will help with preoperative risk stratification, patient counseling, organ allocation optimization, and potentially improving surgical outcomes if the risk factors are modifiable. Our experimental results on the OPTN dataset demonstrate the model's competitive performance in PGD prediction for heart transplant recipients. Notably, our analysis of individual feature attributions and interactions has unveiled novel risk factors potentially contributing to PGD and the interplay between these features. These discoveries shed light on previously overlooked factors and their complex interactions, offering promising avenues for improved PGD risk management in heart transplant patients.

We summarize our contributions as follows:

- We propose a novel ML framework to precisely predict the PGD in heart transplants with the ROC-AUC of 0.868 to support the clinical decision.
- We interpret the prediction results by identifying the salient features in prediction with computational methods and medical expertise.
- Automatically discover the interactions among features between donor and recipient. We can identify the top 20 valuable feature interactions with clinical meanings.

The proposed ML framework still has limitations in predicting heart PGD. We discuss the limitations from data, model, and clinical perspectives. From the data perspective, the patients in the UNOS are the accepted patients of hospitals and transplant centers. However, this could be biased because some patients who visited the hospitals did not get accepted and registered on the waiting list. So, the model is unsuitable for determining the patient's eligibility for heart transplants. Another limitation of the data is the modality of the UNOS data. We only have part of the features collected from patients and donors. Some modalities and more granular patient information are missing in the records provided by UNOS.

In our findings, the important and predictive features for PGD are also highly related to other complications after transplant. Notably, features like DIAB and COD_CAD_DON are associated with heightened systemic inflammation, while hemodynamic parameters (eg, HEMO_PA_MN_TCR) suggest recipient instability may exacerbate inflammatory responses. These discoveries underscore the importance of inflammation in PGD pathogenesis and warrant further investigation into its mechanistic role.

The feature importance ranking highlights key predictors of PGD, including DISTANCE, AGE_DON, and INT_WGT_KG_CALC, which reflect the impact of distance, donor age, and recipient weight on transplant outcomes. Additionally, ABO and ABO_DON highlight blood type compatibility, and LV_EJECT points to donor cardiac function as a critical factor. The inclusion of ETHCAT suggests potential genetic or sociocultural influences on PGD risk. These findings provide actionable insights for optimizing donor selection and reducing PGD, though future research should explore integrating genetic and social data to further refine predictive models.

The feature importance ranking confirms several already known risk factors for PGD, including AGE_DON, DISTANCE, DIAB, COD_CAD_DON, HEMO_PA_MN_TCR, HEMO_SYS_TCR, and LV_EJECT. Additionally, it identifies new potential risk factors, such as INT_WGT_KG_CALC, ETHCAT, INT_HGT_CM_CALC, and BMI_DON_CALC, which warrant further investigation.

Meanwhile, the feature selection is conducted by human experts, which may contain bias and not be optimal. There are also computational algorithms that automatically select features.[22] From the model perspective, we only use the classical model architectures of MLP, which have already achieved competitive prediction accuracy. There are more advanced neural network architectures that could provide more powerful performance. In this work, we did not attach importance to the fairness of the model's output. The model's prediction may be biased towards some subgroups, which can be investigated in future work. From the clinical perspective, we did not compare it with the heuristic method designed by an expert due to the lack of a standard scoring method to measure the PGD in heart transplants. Meanwhile, a comparison between the ML model's prediction performance and that of the human expert has not been conducted on the in-hospital data.

Our ML model demonstrates potential in predicting PGD; however, its accuracy is limited by the absence of key variables such as social and genetic factors. As such, the model is designed to augment, rather than replace, clinical decision-making. Future studies should aim to incorporate these additional data types to enhance predictive performance and build a human-in-the-loop clinical decision support system for PGD.

In the future, it is a promising direction to explore variable combinations and mathematical relationships to improve PGD prediction. Investigating interactions between predictors and leveraging advanced ML models like ensemble learning could discover complex patterns, refine accuracy, and provide actionable insights for optimizing donor selection and post-transplant care. Meanwhile, we will also take other factors into consideration, for example, experience of surgeon.

Conclusion

In this work, we propose a ML workflow to precisely predict the PGD after a heart transplant and identify the high-risk factors related to PGD with feature importance and interacted features. We analyze the results from both computational and clinical perspectives. This work paves a way to apply AI in heart transplant and we plan to explore more of the integration of this method in the real-world clinical workflow.

Author contributions

Sirui Ding designed and implemented the method and the manuscript writing. Chia-Yuan Chang reviewed the

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manuscript and method with suggestions. Yafen Liang and Cheryl Brown helped with writing and provided domain expertise. Xiaoqian Jiang, Xia Hu, and Na Zou helped in the study design and funding acquisition.

Supplementary material

Supplementary material is available at *Journal of the American Medical Informatics Association* online.

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Conflicts of interest

There is nothing to declare.

Data availability

The data is available at UNOS heart transplant database. United Network for Organ Sharing is a nonprofit organization responsible for overseeing the US national organ transplant system. This study utilizes data from the UNOS database, which includes comprehensive records of all transplants conducted in the United States since 1986. Researchers interested in accessing the data can submit a request through: https://unos.org/data/.

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